Course Code	EC24P
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Class	T.E – EXTC SEM-V
Division	B
Date of Performance	22nd August 2025
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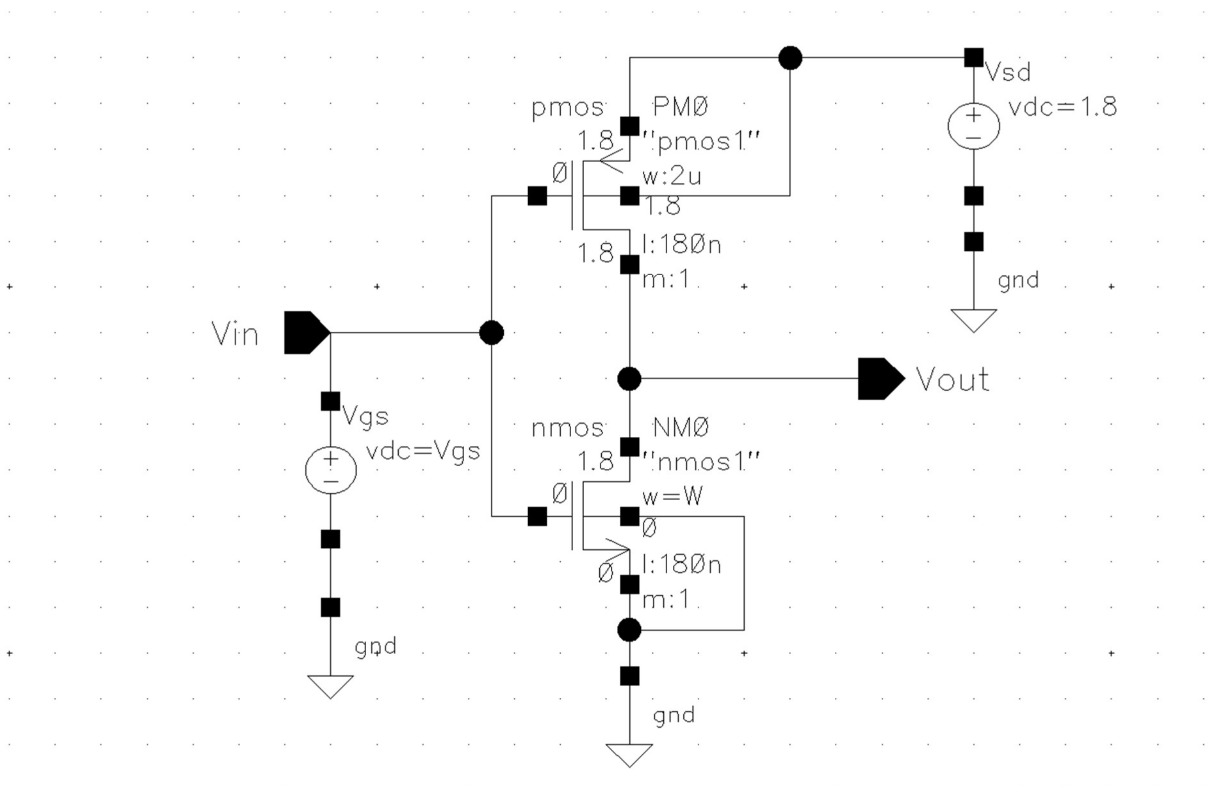
EXPERIMENT NO.5

Estimated time to complete this experiment: 02 hours.
Objective: To observe the effect of varying aspect ratio of NMOS & PMOS on VTC of CMOS inverter.
CO to be achieved: CO2.
Expected Outcome of Experiment: • Students will learn cadence virtuoso. • Students will be able to plot VTC of CMOS inverter using CADENCE Virtuoso. • Students will be able to design CMOS inverter.
Pre Lab/ Prior Concept: Knowledge of working of MOSFET and CMOS Inverter.
Apparatus/TOOL: • PC • CADENCE VIRTUOSO.

Circuit Diagram:



5) Procedure.

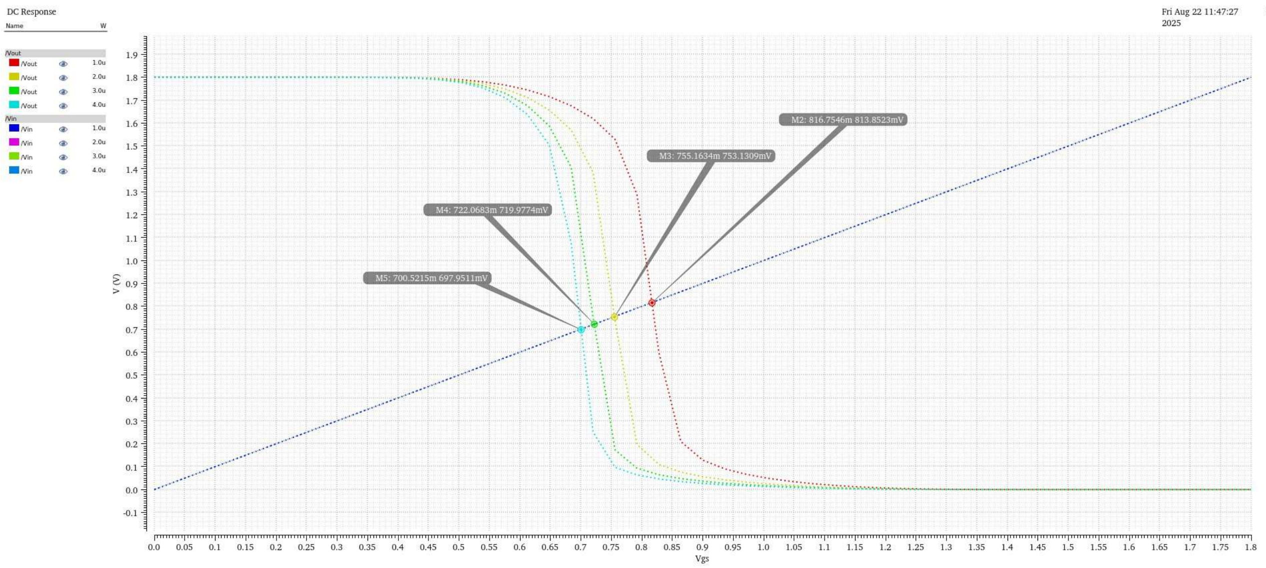
- Open the cadence virtuoso tool in Red-Hat Linux. The commands are
  - csh
  - source /home/install/cshrc
  - virtuoso.
- Create the Library and cell view
- Open log window using terminal.
- Click on the new option, now create library with gpdk180 from technology library.
- Again click the on new option, and set the cell view.
- Cadence Virtuoso dashboard will be opened.

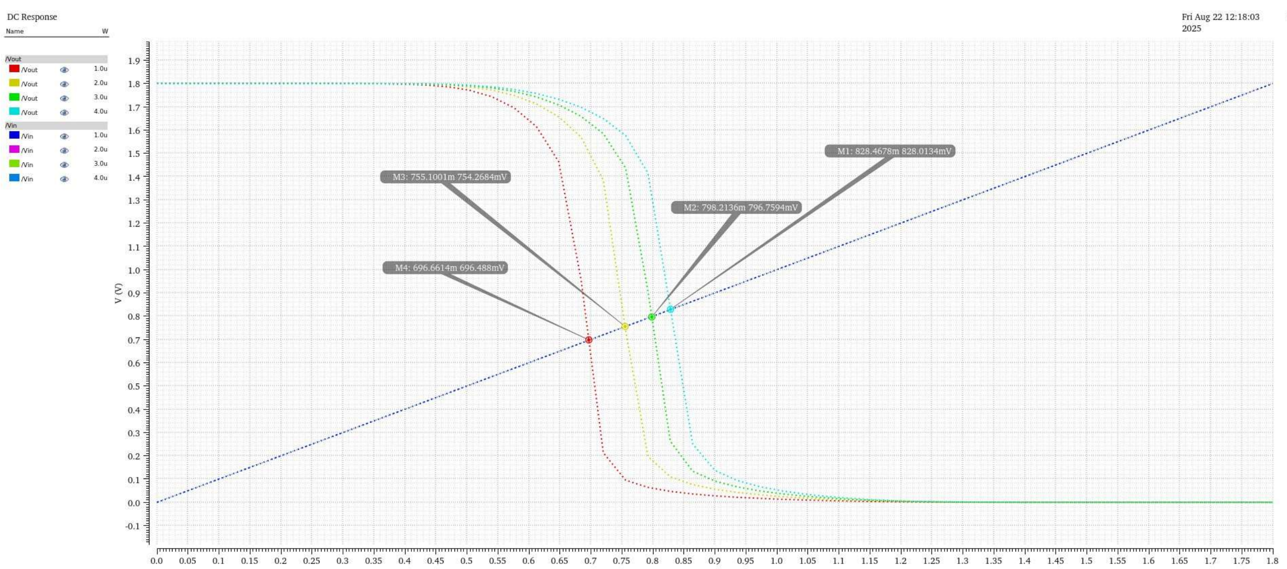
<u>For characterization of 180nm CMOS Inverter.</u>

- Select NMOS transistor from add instance and select its channel length as 180nm (Nano meter) and channel width as W um (micro-meter).
- Select PMOS transistor from add instance and select its channel length as 180nm (Nano meter) and channel width as W 1um (micro-meter).
- Construct the circuit shown in circuit diagram in Cadence.
- Take VDD=1.8V and VGG=vgg V.
- Then plot the Voltage Transfer Characteristics of CMOS Inverter for W=1um to 3um.

- Then find the Threshold voltage of CMOS inverter for each value of width W.
- Now for NMOS transistor take L=180nm and W=1um for PMOS transistor tale L=180nm and width as W um.
- Now plot the VTC of CMOS Inverter by varying width of PMOS from 1um to 3um and find the threshold voltage of CMOS inverter for each width value of PMOS transistor.

Results:





Observation.

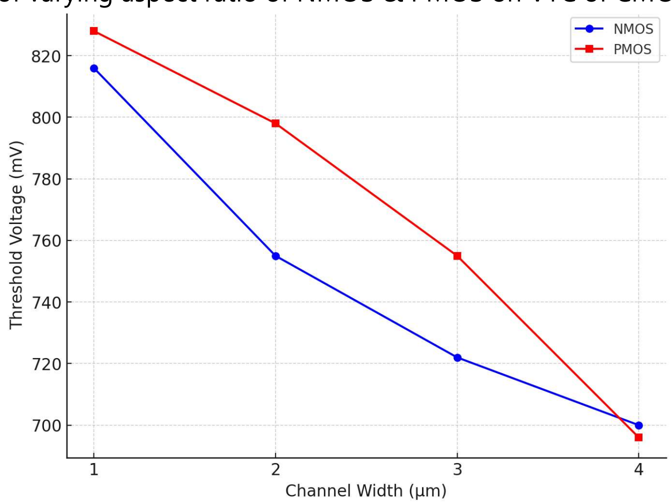
Sr No.	Channel width of NMOS.	Threshold voltage of Inverter.
1	1um	816m
2	2um	755m
3	3um	722m
4	4um	700m
Sr No.	Channel width of PMOS	Threshold voltage of Inverter.
1	1um	828m
2	2um	798m
3	3um	755m
4	4um	696m

Conclusion:

By varying the width (W) of the NMOS and PMOS transistors, we analysed the changes in the Voltage Transfer Characteristics (VTC) of the CMOS inverter.
Increasing the NMOS width made the inverter transition from high to low more quickly and caused the threshold voltage to decrease.
Increasing the PMOS width improved its ability to pull the output high, resulting in a shift of the threshold voltage toward a higher value.

Post lab Questions.

Q1) What is the effect of varying aspect ratio of NMOS & PMOS on VTC of CMOS Inverter?



Practical Performance (5 Marks)	Presentation and Conclusion (5 Marks)	Total (10 Marks)	Sign	Date of Submission.